MSc in Computer Science and Engineering

Planning, Learning and Decision Making

Homework 2: Markov Decision Processes
Smart Home Energy Management

The Setup

Consider a simplified battery with two states: S_0 (Empty) and S_1 (Charged). To provide energy to the house the agent has two actions: **UseGrid** (a_G) and **UseBattery** (a_B) that, based on the weather, might allow to both provide energy and charge the battery. The transition matrices for each action are:

$$P^{a_G} = \begin{bmatrix} 0.3 & 0.7 \\ 0.1 & 0.9 \end{bmatrix}, \quad P^{a_B} = \begin{bmatrix} 0.9 & 0.1 \\ 0.7 & 0.3 \end{bmatrix}$$

The costs are $C(s, a_G) = 10$ for all s, and $C(S_0, a_B) = 50$, $C(S_1, a_B) = 0$. Use $\gamma = 0.9$.

Exercise 1: Bellman Expectation

Assume an initial value estimate $V(S_0) = 100$ and $V(S_1) = 20$. Calculate the action cost to go $Q(S_1, a_G)$ and $Q(S_1, a_B)$. Which action is optimal for this iteration?

$C^\pi = \begin{bmatrix} 10 \\ 10 \end{bmatrix} + 0.9\, P^\pi = \begin{bmatrix} 0.3 & 0.7 \\ 0.1 & 0.9 \end{bmatrix}$ V

Exercise 2: Policy Evaluation

For a fixed policy π that **always** chooses the **Grid** action, solve the system of linear equations $V^\pi = C^\pi + \gamma P^\pi V^\pi$ to find the exact steady-state values for both states.

$V = \begin{bmatrix} 100 \\ 20 \end{bmatrix}$ $C = \begin{bmatrix} 10 & 50 \\ 10 & 0 \end{bmatrix}$ $\pi = \begin{array}{c|cc} & G & B \\ \hline S_0 & 1 & 0 \\ S_1 & 1 & 0 \end{array}$

Exercise 3: Contraction Mappings

The Bellman Optimality Operator T is defined as $(TV)(s) = \min_a [C(s,a) + \gamma \sum_{s'} P(s'|s,a) V(s')]$. $\begin{bmatrix} 100 \\ 20 \end{bmatrix}$

1. Prove that if we add a constant C to our value function such that $V_{new} = V_{old} + C$, the new value after one Bellman update will be $TV_{new} = TV_{old} + \gamma C$.

2. Use this property to explain why Value Iteration is guaranteed to converge to a unique fixed point regardless of the initial V_0.

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